

| Sr. No. | Topic Name                                                                                                                  |
|---------|-----------------------------------------------------------------------------------------------------------------------------|
| 1       | Introduction to Compiler: Phases of Compiler                                                                                |
| 2       | Passes of Compiler                                                                                                          |
| 3       | Cross compiler, Bootstrapping                                                                                               |
| 4       | Finite State Machines and Regular Expressions, along with their applications in lexical analysis.                           |
| 5       | Implementation of lexical analyzers                                                                                         |
| 6       | Lexical Analyzer Generator and the LEX Compiler.                                                                            |
| 7       | Formal grammars and their application to syntax analysis                                                                    |
| 8       | BNF notation and ambiguity in context-free grammars.                                                                        |
| 9       | The syntactic specification of programming languages: Context free grammars                                                 |
| 10      | Derivations and parse trees, and the expressive capability of Context-Free Grammars (CFGs).                                 |
| 11      | Basic Parsing Techniques: Fundamental parsing techniques and types of parsers.                                              |
| 12      | Shift reduce parsing                                                                                                        |
| 13      | Operator precedence parsing                                                                                                 |
| 14      | Top-down parsing techniques.First and Follow set                                                                            |
| 15      | Predictive parsers                                                                                                          |
| 16      | Automatic Construction of efficient Parsers: LR parsers                                                                     |
| 17      | The canonical Collection of LR(0) items                                                                                     |
| 18      | Constructing SLR parsing tables                                                                                             |
| 19      | Construction of canonical LR parsing                                                                                        |
| 20      | Constructing LALR parser,YACC                                                                                               |
| 21      | Constructing LALR parsing Table                                                                                             |
| 22      | An automatic parser generator                                                                                               |
| 23      | Implementation of LR parsing Table                                                                                          |
| 24      | Syntax-directed Definition& Translation: Syntax- directed Definition                                                        |
| 25      | Implementation of Syntax- directed Translators                                                                              |
| 26      | Intermediate code, postfix notation                                                                                         |
| 27      | Syntax trees & DAG                                                                                                          |
| 28      | Three address code, quadruple, triples and Indirect Triples                                                                 |
| 29      | Translation of assignment statements, Boolean expressions                                                                   |
| 30      | Statements that alter the flow of control, postfix translation, and translation with a top-down parser.                     |
| 31      | More about translation: Array references in arithmetic expressions                                                          |
| 32      | Declaration and case statements                                                                                             |
| 33      | Symbol Tables&Code Generation: Data structure for symbols tables                                                            |
| 34      | Run-Time Administration: Implementation of simple stack allocation scheme, storage allocation in block structured language. |
| 35      | Error Detection & Recovery, Lexical Phase errors, syntactic phase errors, semantic errors                                   |
| 36      | Design Issues, the Target Language,Addresses in the Target Code                                                             |
| 37      | Optimization of Basic Blocks, Code Generator,Basic Blocks and Flow Graphs                                                   |
| 38      | Code optimization: Machine-Independent Optimizations, Loop optimization                                                     |
| 39      | DAG representation of basic blocks,Global Data-Flow analysis.                                                               |